Synthetic Fixed-Confidence Experiments for Binary Pairwise Pareto Set Identification

Experiment report generated from `Exp_Synthetic`

July 13, 2026

Abstract

This report summarizes the completed synthetic experiments for fixed-confidence binary pairwise Pareto set identification. The experiments compare VB-EGE-practical against three fixed-confidence benchmarks: focal Borda, BT-MLE-Cert, and pairwise Borda plug-in. The MLE method uses a fixed-confidence-style empirical certificate and is not claimed to have a formal delta-correct confidence region. The main runs contain 3400 scaling replications and 2700 benchmark replications per algorithm across all experiment cells. Across all completed main settings, the empirical error rate is 0.0 for scaling and 0.0 for benchmarks; therefore the main comparison is sample complexity, with median stopping time, paired ratios, and quantiles primary and raw means secondary. The experiments are organized in logical order: fixed-confidence scaling, confidence-scaling quantiles, the fixed-confidence benchmark baseline suite, constant sensitivity, Pareto-size ablation, correlated arena diagnostics, sampling-mechanism diagnostics, and fixed-error budget calibration.

1 Protocol

The task is fixed-confidence identification of the strict Pareto set under binary pairwise comparison feedback. Each run stops adaptively when the algorithm can return a Pareto set certificate at the target failure probability. The default target is $\delta = 0.05$, except in the confidence-scaling sweep where δ itself is varied. The reported stopping time counts coordinate-specific pairwise comparison pulls. Lower stopping time is better when the empirical error rate is controlled.

Table 1: Algorithms compared with fixed-confidence stopping interfaces. The MLE row is an empirical certificate heuristic, as detailed below.

Name	Sampling model	Fixed-confidence rule
VB-EGE-practical	Coordinate-wise Vector-Borda estimates	Adaptive accept/reject using empirical gap certificates
UniformFocalBorda-FC	Uniform coordinate-wise Borda sampling	Same confidence threshold family without adaptive elimination
UniformPairwiseBT-MLE-Cert	Uniform pair-coordinate Bradley–Terry sampling with MLE	Empirical plug-in certificate; no formal MLE confidence region
UniformPairwiseBT-BordaPlugIn-FC	Uniform pair-coordinate sampling with Borda plug-in	Stop when Borda plug-in certificate is satisfied

Definitions used by the report. There are K arms and d objectives. A coordinate-specific comparison between arms i and j on objective r follows a Bradley–Terry model,

$$Y_{ijr} \sim \text{Bernoulli}(p_{ijr}), \qquad p_{ijr} = \sigma(\theta_{ir} - \theta_{jr}).$$

The coordinate-wise Borda embedding is

$$B_{ir} = \frac{1}{K-1}\sum_{j\neq i} p_{ijr}.$$

Most certificates in this report are computed in this Vector-Borda space. The synthetic generators used for the main experiments pass a Borda/latent Pareto consistency check, so the reported true Pareto set agrees under θ and under B.

For any score matrix $X \in \mathbb{R}^{K\times d}$, define

$$m_{ij}(X) = \min_r (X_{jr} - X_{ir}), \qquad M_{ij}(X) = \max_r (X_{ir} - X_{jr}), \qquad \Delta_i^\star(X) = \max_{j\neq i} m_{ij}(X).$$

If arm i is non-Pareto, its identification gap is $\Delta_i(X) = \Delta_i^\star(X)$. If arm i is Pareto, the reported gap is

$$\Delta_i(X) = \min_{j\neq i} \min\{M_{ij}(X), \max(M_{ji}(X), 0) + \max(\Delta_j^\star(X), 0)\}.$$

The minimum Borda gap is $\Delta_{\min,B} = \min_i \Delta_i(B)$. The Borda hardness is

$$H_B = \sum_i \Delta_i(B)^{-2},$$

with $H_B = \infty$ when a gap is non-positive or non-finite. Large H_B means that one or more arms lie close to the Pareto decision boundary; many plots therefore also report $\tau/(dH_B)$ to remove first-order instance hardness.

All fixed-confidence methods use the same practical constants: $c_{\text{samp}} = 2$, $c_{\text{thr}} = 4$, and $c_{\log} = 4$. In phase m, with cell count C, the schedule is

$$\varepsilon_m = 2^{-m}, \qquad L_m = \log(c_{\log} C m^2/\delta), \qquad n_m = \left\lceil c_{\text{samp}} L_m/\varepsilon_m^2 \right\rceil, \qquad r_m = \sqrt{L_m/(2n_m)}.$$

The sample constant c_{samp} multiplies the per-cell sample count, the log constant $c_{\log}$ is the union-bound safety factor inside L_m, and the threshold constant c_{thr} is the certificate margin. A fixed-confidence baseline stops when its estimated minimum gap satisfies $\widehat{\Delta}_{\min} > c_{\text{thr}} r_m$.

Benchmark implementations. The three benchmarks other than VB-EGE are implemented as follows.

- **UniformFocalBorda-FC.** In each phase it samples every arm-objective cell (i, r) up to n_m Borda observations. Equivalently, the simulation draws $S_{ir} \sim \text{Binomial}(N_{ir}, B_{ir})$ and estimates $\widehat{B}_{ir} = S_{ir}/N_{ir}$. It computes the plug-in strict Pareto set from $\widehat{B}$, computes $\widehat{\Delta}_{\min}$ using the gap formula above, and stops when $\widehat{\Delta}_{\min} > c_{\text{thr}} r_m$. It does not eliminate arms adaptively; it keeps all Kd cells balanced.

- **UniformPairwiseBT-MLE-Cert.** In each phase it samples every pair-coordinate cell (i, j, r), $i < j$, up to n_m comparisons, so $C = dK(K-1)/2$. For each objective r, it fits a separate ridge-stabilized BT MLE from the win counts:

$$\min_{\vartheta_{\cdot r}} \sum_{i<j} \{W_{ijr} \log(1 + \exp[-(\vartheta_{ir} - \vartheta_{jr})]) + W_{jir} \log(1 + \exp[\vartheta_{ir} - \vartheta_{jr}])\} + \frac{\lambda}{2}\|\vartheta_{\cdot r}\|_2^2.$$

The fitted coordinate scores are centered for identifiability. The algorithm then computes the plug-in Pareto set and $\widehat{\Delta}_{\min}$ from $\widehat{\theta}$ and stops when the same gap certificate is satisfied. If the graph is disconnected, the optimizer fails, or the estimate is too large, the implementation reruns the fit with the fallback ridge value. This method is labeled "Cert" rather than "FC" because the Bernoulli cell radius r_m is not, by itself, a valid uniform confidence radius for $\|\widehat{\theta}-\theta\|_\infty$. A formal BT-MLE guarantee would also need likelihood curvature, the comparison-graph Laplacian, dynamic range, ridge, and gauge fixing. Thus this baseline has a fixed-confidence-style stopping interface but is an empirical certificate heuristic; zero observed error is not presented as a proof of δ-correctness.

- **UniformPairwiseBT-BordaPlugIn-FC.** It uses the same balanced pair-coordinate sampling as the MLE baseline, but avoids solving the BT MLE. It estimates each pairwise probability with add-1/2 smoothing,
$$\widehat{p}_{ijr} = \frac{W_{ijr} + 1/2}{N_{ijr} + 1},$$
forms the plug-in Borda estimate $\widehat{B}_{ir} = (K-1)^{-1} \sum_{j \neq i} \widehat{p}_{ijr}$, and applies the same strict-Pareto plug-in set and gap certificate in Borda space.

In the benchmark runs, the BT-MLE baseline uses balanced random pair-coordinate allocation, ridge parameter 10^{-8}, fallback ridge parameter 10^{-4}, maximum absolute latent score 20, and MLE tolerance 10^{-9} in the benchmark run. Raw outputs are stored as CSV files because the local parquet engine was unavailable; this does not change the numerical results.

Metrics and legends. The primary metrics are median stopping time and paired per-replication ratios. Raw mean stopping time $\bar{\tau}$ is retained for symmetric scaling laws, where the distribution is well behaved. The normalized budget $\tau/(dH_B)$ uses the instance Borda hardness H_B and is used only to put settings with different dimensions and gaps on a comparable axis. Across the main fixed-confidence runs the empirical Pareto-set error is zero; this is reported with Wilson upper bounds rather than uninformative zero-error floor curves. The formal baselines in the report are the fixed-confidence baselines, not capped fixed-budget variants. The fixed-error budget calibration uses capped variants only to show what happens before certification. In line plots, the legend entries are the four algorithm names. In bar plots, the x-axis labels are algorithm names and the y-axis is log-scaled median stopping time with interquartile error bars.

Plot computation conventions. Whenever the same axis label appears later, it uses the same computation. Mean stopping time is $\bar{\tau} = n^{-1} \sum_s \tau_s$ over replications. Median, q25, q75, q90, and q95 stopping times are empirical quantiles of the same $\{\tau_s\}$ values. Empirical error is $n^{-1} \sum_s \mathbf{1}\{\widehat{P}_s \neq P_s\}$; when no failures are observed, error-floor plots show $\log_{10}(0.5/n)$ instead of $-\infty$. The Wilson upper bound is the upper endpoint of the two-sided 95% Wilson interval for this binomial error rate. A main-benchmark paired ratio first computes $R_s = \tau_{\text{baseline},s}/\tau_{\text{VB},s}$ on the same instance and observation replicate, then reports $\text{median}_s\, R_s$ and a percentile 95% bootstrap interval for that median. Where hierarchical instance IDs are available, latent instances are resampled as clusters with all of their observation replications; legacy fixed-instance cells use replication bootstrap. Other ratio plots use the explicitly stated ratio of summary statistics. Pair-cell coverage is averaged over runs after computing τ_s/C for each pairwise run, where $C = dK(K-1)/2$. Mean Hamming distance is the average symmetric-difference size $|\widehat{P}_s \triangle P_s|$. The fixed-error budget ratio is $T/\widetilde{\tau}_{\text{VB,ref}}$, where $\widetilde{\tau}_{\text{VB,ref}}$ is the default fixed-confidence VB-EGE median stopping time on the matching benchmark setting. Final phase is the last dyadic phase index m used by the algorithm. Achieved objective correlation is the mean off-diagonal sample correlation among objective columns of the generated instance.

2 Fixed-Confidence Scaling

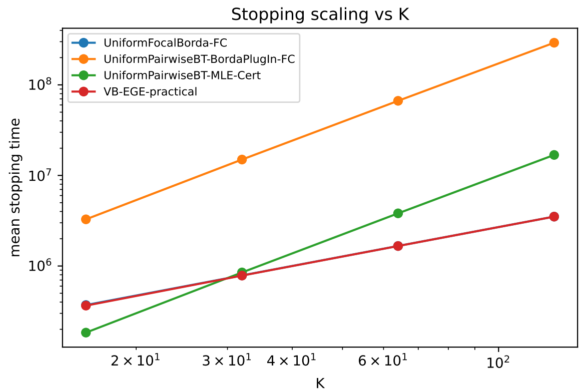
Table 2: Fixed-confidence scaling settings. All tasks use the symmetric hard generator and four algorithms; each cell is averaged over 200 random permutations/seeds.

Sweep	Generator	Grid	Purpose
K scaling	symmetric_hard	K in {16,32,64,128}; d=4; Delta=1; delta=0.05	One Pareto arm at the origin and K-1 dominated arms at -Delta in every coordinate.
d scaling	symmetric_hard	K=64; d in {2,4,10}; Delta=1; delta=0.05	Symmetric hard instance with the number of dimensions varied.
Gap scaling	symmetric_hard	K=64; d=10; Delta in {0.5,0.75,1,1.25,1.5}; delta=0.05	Symmetric hard instance with the latent separation Delta varied.
Confidence scaling	symmetric_hard	K=64; d=10; Delta=1; delta in {0.20,0.10,0.05,0.02,0.01}	Symmetric hard instance with the target failure probability varied.

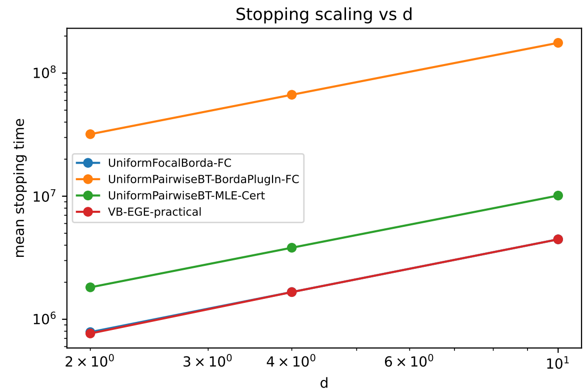
Table 3: Scaling results. For each swept parameter value, the table reports VB-EGE mean stopping time and baseline ratios relative to VB-EGE. All empirical error rates are zero.

Sweep	Value	VB-EGE $\bar{\tau}$	Focal/VB	MLE/VB	Plug-in/VB
K scaling	16	3.65×10^5	1.02x	0.50x	8.97x
K scaling	32	7.84×10^5	1.00x	1.08x	19.1x
K scaling	64	1.66×10^6	1.00x	2.29x	40.1x
K scaling	128	3.51×10^6	1.00x	4.80x	83.2x
d scaling	2	7.67×10^5	1.03x	2.37x	41.6x
d scaling	4	1.66×10^6	1.00x	2.29x	40.1x
d scaling	10	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	0.50	1.84×10^7	1.00x	2.31x	39.2x
Gap scaling	0.75	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	1	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	1.25	1.82×10^6	2.44x	1.27x	23.4x
Gap scaling	1.50	1.07×10^6	1.00x	2.17x	39.8x
Confidence scaling	0.01	4.99×10^6	1.00x	2.24x	38.6x
Confidence scaling	0.02	4.76×10^6	1.00x	2.25x	39.0x
Confidence scaling	0.05	4.46×10^6	1.00x	2.27x	39.5x
Confidence scaling	0.10	4.23×10^6	1.00x	2.29x	39.9x
Confidence scaling	0.20	4.01×10^6	1.00x	2.30x	40.4x

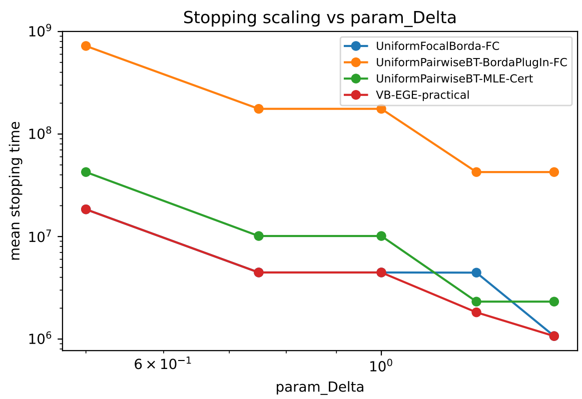
The scaling tasks isolate how stopping time changes with the number of arms, the dimension, the latent gap, and the confidence level on the symmetric hard instance. In this generator, there is a single true Pareto arm and all other arms are dominated by the same coordinate gap. This makes the setting useful for checking whether the fixed-confidence rules behave monotonically and whether pairwise baselines pay a large cost for broad pair-coordinate coverage.



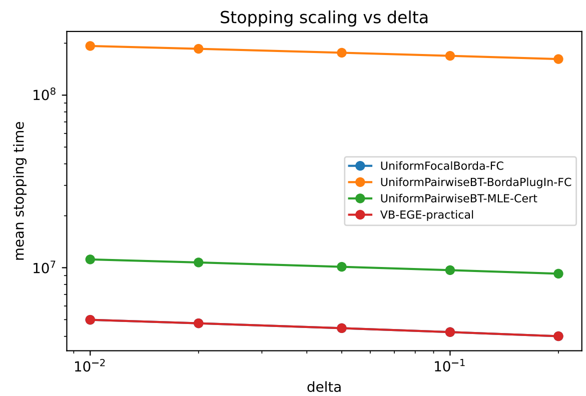
(a) K scaling



(b) d scaling



(c) Gap scaling



(d) Confidence scaling

Figure 1: Scaling curves for mean stopping time. In each panel, the x-axis is the swept design parameter shown in the subcaption; the y-axis is mean stopping time $\bar{\tau}$ over replications. Both axes are log-scaled when positive, and lower curves indicate fewer comparison pulls.

Result interpretation. Descriptive log–log fits give slopes about 1.09 for both the K sweep and the d sweep, supporting the predicted near-linear leading dependence. VB-EGE-practical and UniformFocalBorda-FC are nearly tied on the symmetric hard instance, as expected because the instance has a single symmetric Pareto arm and little need for adaptive frontier structure. Both methods estimate the same coordinate-wise Vector-Borda cells, and the symmetric generator gives all dominated arms comparable gaps. Therefore most certificates mature in the same final phase, leaving little heterogeneity for VB-EGE's adaptive elimination to exploit. In this setting the near-overlap between the UniformFocalBorda-FC line and the VB-EGE-practical line is a sanity-check outcome rather than evidence against VB-EGE; the method's advantage appears when the instance has richer frontier structure or when pairwise baselines must pay broad pair-coordinate coverage costs. The pairwise BT-MLE and Borda plug-in baselines are substantially slower as K and d grow, reflecting the cost of spreading samples across many pair-coordinate cells. Larger gaps reduce stopping time, while smaller target delta increases stopping time through the confidence threshold. The small plateaus in the gap sweep are consistent with dyadic phase schedules: once two gap values trigger the same final phase, their stopping times can match even if the underlying gap differs.

3 Confidence-Scaling Quantiles

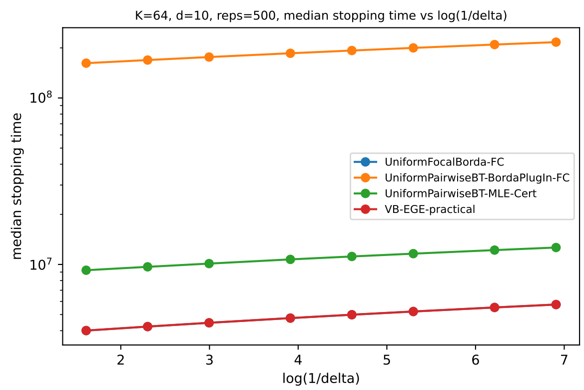
Table 4: Confidence-scaling quantile settings. These runs use the same fixed-confidence algorithms as the main benchmark suite and vary only the requested failure probability.

Setting	Generator	Grid	Purpose	Reps
Symmetric K64 d10	symmetric_hard	K=64; d=10; Delta=1; delta in {0.20,0.10,0.05,0.02,0.01,0.005,0.002,0.001}	Stress the tail of fixed-confidence stopping times as log(1/delta) grows.	500
Arena-10 medium	arena_tradeoff_frontier	K=64; d=10; s=16; margins [0.06,0.20]; delta in {0.20,0.10,0.05,0.02,0.01,0.005}	Check confidence scaling on a high-dimensional multi-Pareto arena.	300

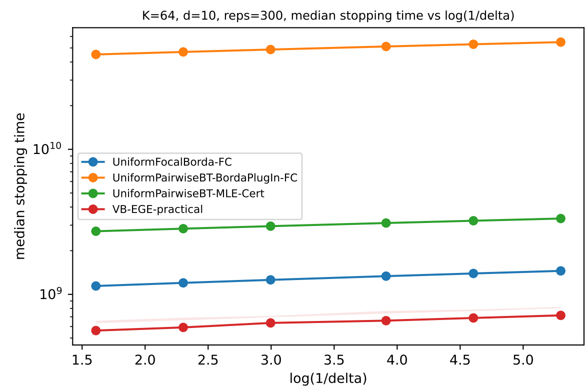
Table 5: Confidence-scaling quantile results. VB columns report mean and 95th-percentile stopping times; ratios use mean stopping time relative to VB-EGE.

Setting	δ	VB mean τ	VB q95 τ	Focal/VB	MLE/VB	Plug-in/VB
Symmetric K64 d10	0.200	4.01×10^6	4.01×10^6	1.00x	2.30x	40.4x
Symmetric K64 d10	0.100	4.23×10^6	4.23×10^6	1.00x	2.29x	39.9x
Symmetric K64 d10	0.050	4.46×10^6	4.46×10^6	1.00x	2.27x	39.5x
Symmetric K64 d10	0.020	4.76×10^6	4.76×10^6	1.00x	2.25x	39.0x
Symmetric K64 d10	0.010	4.99×10^6	4.99×10^6	1.00x	2.24x	38.6x
Symmetric K64 d10	0.005	5.22×10^6	5.22×10^6	1.00x	2.23x	38.3x
Symmetric K64 d10	0.002	5.52×10^6	5.52×10^6	1.00x	2.21x	38.0x
Symmetric K64 d10	0.001	5.74×10^6	5.74×10^6	1.00x	2.20x	37.7x
Arena-10 medium	0.200	5.62×10^8	6.57×10^8	2.09x	4.99x	82.7x
Arena-10 medium	0.100	5.89×10^8	6.91×10^8	2.10x	5.01x	81.8x
Arena-10 medium	0.050	6.21×10^8	7.10×10^8	2.09x	4.95x	80.9x
Arena-10 medium	0.020	6.49×10^8	7.69×10^8	2.12x	4.93x	81.1x
Arena-10 medium	0.010	6.81×10^8	7.86×10^8	2.13x	4.92x	80.2x
Arena-10 medium	0.005	7.06×10^8	8.20×10^8	2.12x	4.87x	79.9x

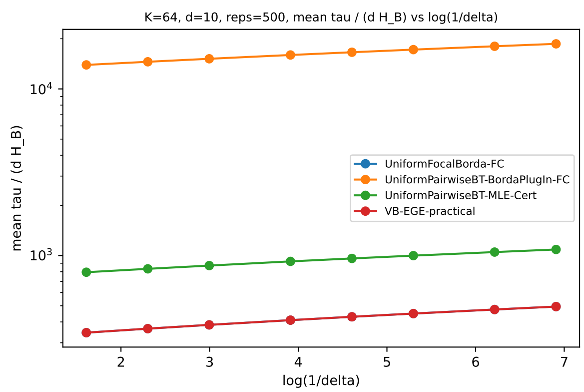
This experiment repeats fixed-confidence runs over a wider grid of target failure probabilities. Within each experiment and replication, all δ values reuse the same synthetic instance, so the confidence effect is not confounded with between-instance hardness variation. The line legends are the four algorithms; the quantile traces report the median and upper stopping-time quantiles. The x-axis is $\log(1/\delta)$, so approximately linear growth indicates the expected logarithmic confidence cost. The normalized panels divide by the instance Borda hardness, and the $\tau/\log(1/\delta)$ panels make deviations from pure logarithmic scaling visible.



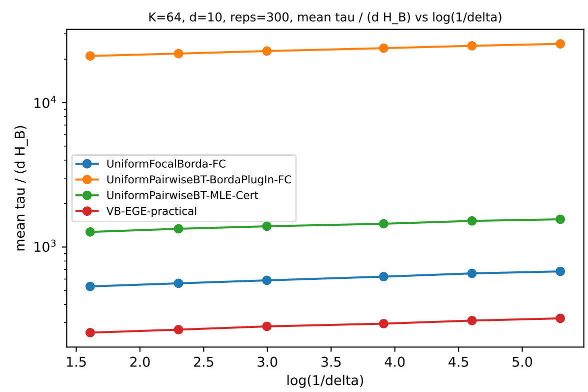
(a) Symmetric K64 d10



(b) Arena-10 medium

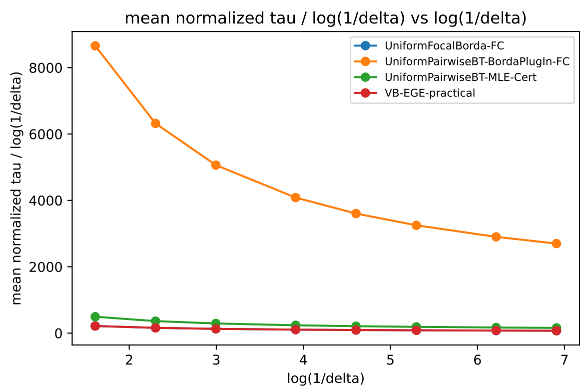


(c) Normalized symmetric

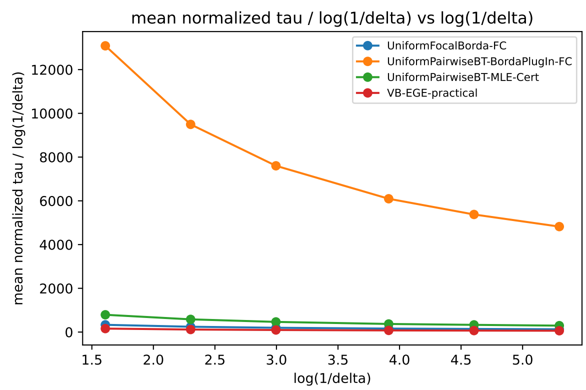


(d) Normalized arena10

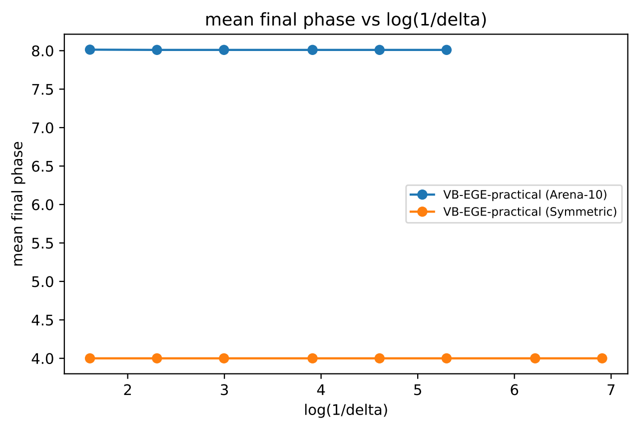
Figure 2: Confidence-scaling stopping quantiles. The x-axis is $\log(1/\delta)$. Non-normalized panels use median τ on the y-axis with q90–q95 shaded bands; normalized panels use mean $\tau/(dH_B)$. Each color is one algorithm.



(a) Symmetric tau/log



(b) Arena10 tau/log



(c) VB final phase

Figure 3: Confidence-scaling diagnostics. The tau/log panels plot mean $\tau/\log(1/\delta)$ against $\log(1/\delta)$. The final-phase panel plots mean final phase index m against $\log(1/\delta)$, with separate lines for the symmetric and Arena settings.

Result interpretation. On the symmetric setting, VB-EGE and the focal Borda baseline remain close because the certificate is essentially one-dimensional after symmetry. The arena setting separates the methods more clearly: VB-EGE keeps a smaller high quantile stopping time than the pairwise baselines because it does not need broad pair-coordinate coverage at every confidence level. The final-phase diagnostic should move in steps rather than smoothly, which is expected from phase doubling. A direct fit $\bar{\tau} = a + b\log(1/\delta)$ gives $R^2 \approx 0.99997$ on the symmetric task and $R^2 \approx 0.9978$ on Arena-10. This supports logarithmic confidence cost in stopping time; it does not empirically validate a 0.001 error probability from only 500 replications.

4 Fixed-Confidence Benchmark Suite

Table 6: Fixed-confidence benchmark settings. All algorithms use target failure probability $\delta = 0.05$; $|P|$ denotes the true Pareto set size.

Setting	Generator	K	d	$\lvert P\rvert$	Reps	Parameters
Convex-2D	convex_frontier_2d	60	2	15	500	s=15
Arena-4 small	arena_tradeoff_frontier	32	4	8	500	s=8, margins [0.08,0.25], alpha=0.7
Arena-4 medium	arena_tradeoff_frontier	64	4	12	300	s=12, margins [0.08,0.25], alpha=0.7
Arena-10 medium	arena_tradeoff_frontier	64	10	16	300	s=16, margins [0.06,0.20], alpha=0.7
Witness-4	unique_witness_d	40	4	8	500	s=8, q=4, margins [0.04,0.16]
Witness-10	unique_witness_d	80	10	16	300	s=16, q=4, margins [0.035,0.14]
Two-group-10	highdim_two_group	64	10	15.79	300	K_low=48, K_high=16

Table 7: Benchmark results. VB-EGE is reported by median stopping time and interquartile range. Each baseline entry is the median paired per-replication ratio to VB-EGE with a 95% bootstrap interval. For the hierarchical Arena-10 rerun, latent instances and all of their observation replications are resampled as clusters; legacy fixed-instance cells use replication bootstrap. All empirical error rates are zero.

Setting	VB median τ [IQR]	Focal/VB	MLE/VB	Plug-in/VB	Wilson upper
Convex-2D	1.14×10^8 [1.06×10^8, 1.32×10^8]	2.00x [1.97x, 2.05x]	4.38x [4.30x, 4.41x]	72.8x [71.9x, 73.9x]	0.0076
Arena-4 small	1.17×10^8 [6.46×10^7, 3.38×10^8]	8.50x [8.09x, 9.00x]	9.19x [8.59x, 9.71x]	157x [151x, 166x]	0.0076
Arena-4 medium	4.75×10^8 [2.22×10^8, 2.33×10^9]	18.7x [16.4x, 19.6x]	43.2x [37.2x, 45.9x]	735x [644x, 769x]	0.0126
Arena-10 medium	6.21×10^8 [5.76×10^8, 6.65×10^8]	2.03x [1.98x, 2.08x]	4.76x [4.65x, 4.87x]	78.6x [76.7x, 80.5x]	0.0126
Witness-4	2.30×10^8 [2.20×10^8, 2.37×10^8]	1.24x [1.23x, 1.26x]	1.78x [1.77x, 1.81x]	29.4x [29.3x, 30.0x]	0.0076
Witness-10	1.58×10^9 [1.55×10^9, 1.67×10^9]	1.03x [1.03x, 1.04x]	3.00x [3.00x, 3.03x]	49.5x [49.5x, 50.0x]	0.0126
Two-group-10	9.43×10^8 [4.58×10^8, 3.00×10^9]	13.7x [12.4x, 16.2x]	31.8x [29.1x, 38.2x]	559x [486x, 645x]	0.0126

The benchmark suite stresses different Pareto geometries beyond the symmetric hard case. Convex-2D tests a clean low-dimensional trade-off frontier. Arena settings use mutually non-dominated anchors and dominated alternatives offset from a random anchor. Witness settings create dominated arms that are certified by a unique Pareto witness, which tests whether algorithms can avoid unnecessary global pairwise coverage. Two-group-10 is a high-dimensional separation task with many low-quality arms and a smaller high-quality group.

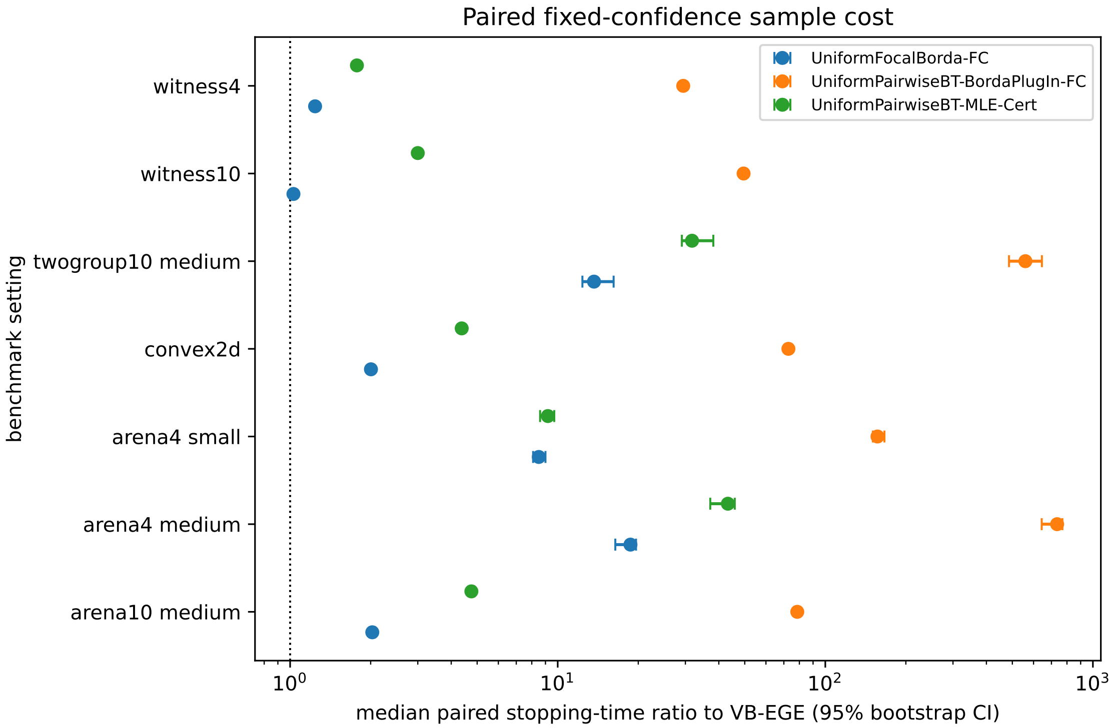
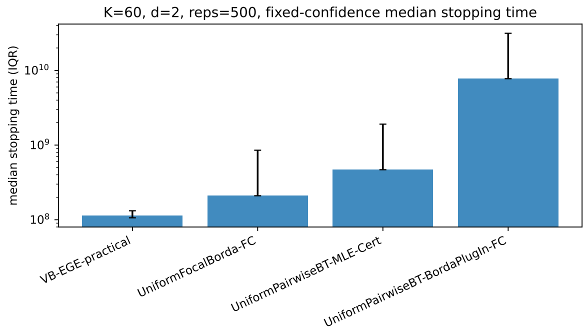
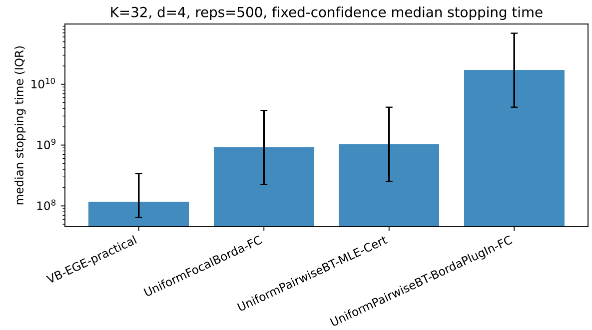


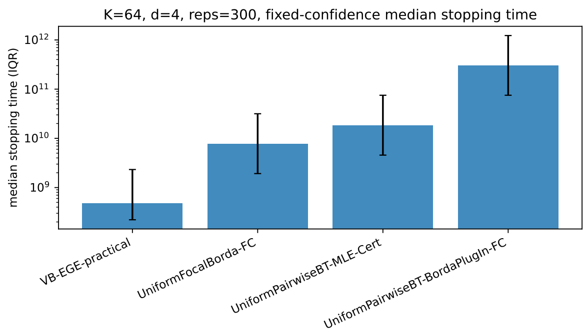
Figure 4: Paired benchmark stopping-time ratios. Each point is $\text{median}_s(\tau_{\text{baseline},s}/\tau_{\text{VB},s})$ on matched replications; horizontal bars are 95% bootstrap intervals for the median. Hierarchical Arena-10 cells resample latent instances as clusters, while legacy fixed-instance cells resample replications. The vertical line at one denotes equal stopping time.



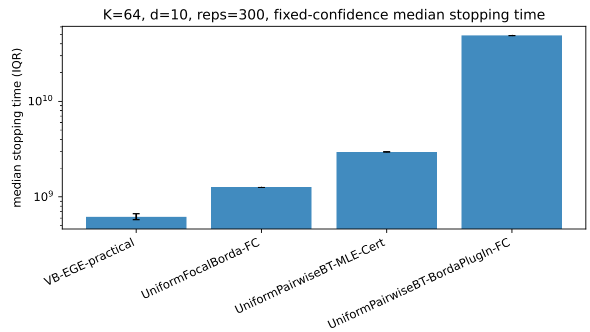
(a) Convex-2D



(b) Arena-4 small

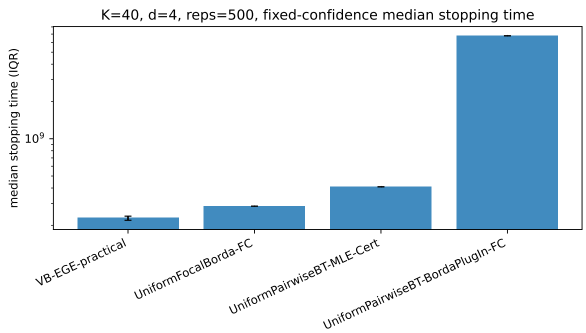


(c) Arena-4 medium

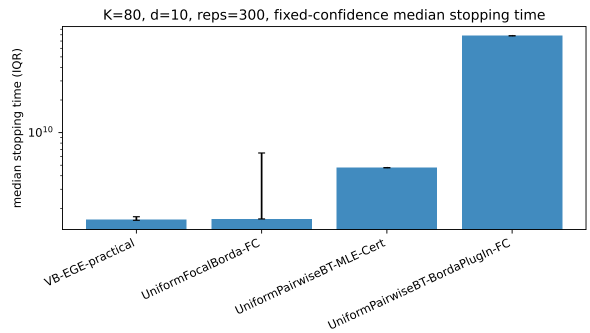


(d) Arena-10 medium

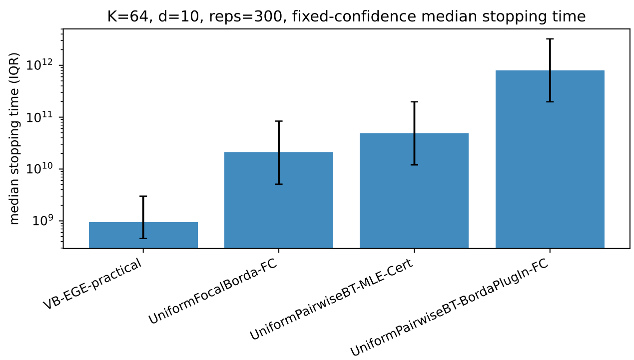
Figure 5: Benchmark stopping times, first group. The x-axis lists algorithms within each benchmark setting; the y-axis is median stopping time on a log scale and error bars span q25–q75. Lower bars mean fewer comparison pulls before certification.



(a) Witness-4



(b) Witness-10



(c) Two-group-10

Figure 6: Benchmark stopping times, second group. The x-axis lists algorithms within each benchmark setting; the y-axis is median stopping time on a log scale and error bars span q25–q75. These tasks emphasize local witnesses and high-dimensional group separation.

Result interpretation. VB-EGE-practical has the smallest median stopping time in every benchmark setting while maintaining zero empirical error. The advantage is moderate on unique-witness instances, where the focal Borda baseline already receives local certificate information, and much larger on arena and two-group instances. The BT-MLE baseline has near-perfect pairwise sign accuracy in the completed runs, but it pays a large exploration cost before enough pair-coordinate cells are covered. The Borda plug-in pairwise baseline is consistently the slowest, which suggests that uniform pairwise sampling is poorly matched to fixed-confidence Pareto certification when the certificate can be expressed in the Vector-Borda embedding.

5 Constant Sensitivity and Theory-Constant Sanity

Table 8: Arena-10 cross-section audit. The benchmark and constant-sensitivity rows use shared bank `arena10_medium_v2`: 30 fixed θ instances with 10 observation replications each and common random numbers. The confidence row retains its original paired-δ bank.

Section	Reps	Mean τ	Median τ	q95 τ
Main benchmark	300	6.13×10^8	6.21×10^8	7.10×10^8
Confidence sweep, $\delta = 0.05$	300	6.21×10^8	6.35×10^8	7.10×10^8
Constant sensitivity, $(2, 4)$	300	6.13×10^8	6.21×10^8	7.10×10^8

The earlier cross-section discrepancy came from independent random instance banks: rare near-boundary Arena instances created very large stopping-time outliers, so raw means differed by orders of magnitude even though medians were similar. The revised benchmark and constant-sensitivity rows now use the same 30 instances, ten observation replications, and common random numbers, so their default results agree exactly. The confidence sweep keeps its original independent bank because its within-replication pairing across δ is the design needed for the confidence-scaling claim; its median remains directly comparable, while its raw mean is not treated as an exact cross-section identity.

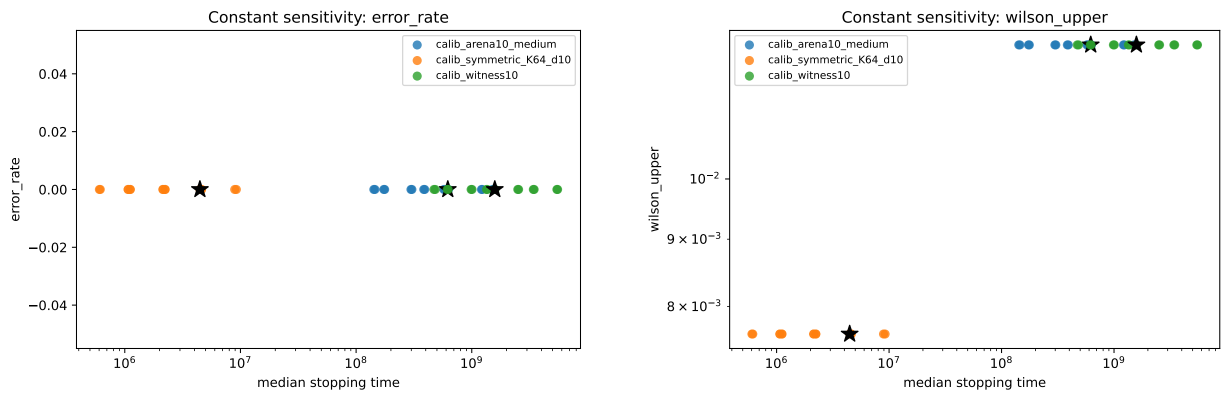
Table 9: Constant-sensitivity settings. VB-EGE is swept over sample constants $\{0.5, 1, 2, 4\}$ and threshold constants $\{2, 3, 4, 6\}$ at $\delta = 0.05$; the focal and BT-MLE-Cert references use the default constants $(2, 4)$.

Setting	Generator	Parameters	Grid	Reps
Symmetric K64 d10	symmetric_hard	K=64; d=10; Delta=1	$c_s \in \{0.5, 1, 2, 4\}$; $c_\theta \in \{2, 3, 4, 6\}$	500
Arena-10 medium	arena_tradeoff_frontier	K=64; d=10; s=16; margins [0.06,0.20]	$c_s \in \{0.5, 1, 2, 4\}$; $c_\theta \in \{2, 3, 4, 6\}$	300
Witness-10	unique_witness_d	K=80; d=10; s=16; q=4; margins [0.035,0.14]	$c_s \in \{0.5, 1, 2, 4\}$; $c_\theta \in \{2, 3, 4, 6\}$	300

Table 10: Constant-sensitivity results. The default row uses $(c_s, c_\theta) = (2, 4)$. The fastest row is the smallest median stopping time among grid points with empirical error at most 0.05 and complete stopping. Focal/default and MLE/default are ratios of median stopping times. Because every tested point has zero observed error, this is sensitivity analysis rather than an identified efficiency–reliability frontier.

Setting	Default VB median τ	Default err	Fastest constants	Fastest median τ	Focal/default	MLE/default
Symmetric K64 d10	4.46×10^6	0.000	(4, 2)	6.03×10^5	1.00x	2.27x
Arena-10 medium	6.21×10^8	0.000	(4, 2)	1.42×10^8	2.03x	4.76x
Witness-10	1.58×10^9	0.000	(2, 2)	4.71×10^8	1.01x	3.00x

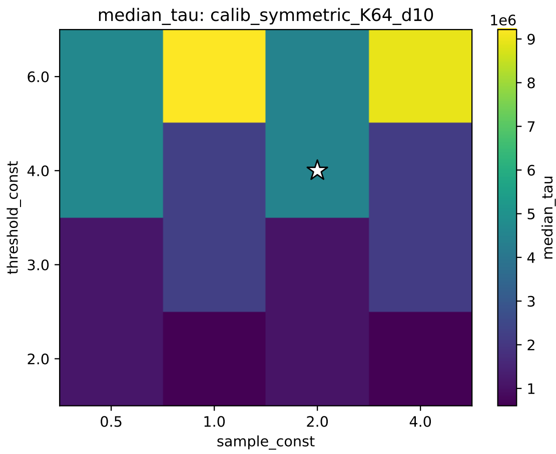
The sensitivity sweep varies the VB-EGE sample constant c_s and threshold constant c_θ. In the sensitivity figures, each point is one constant pair; the x-axis is median stopping time and the y-axis is either empirical error or Wilson upper confidence bound. Heatmaps use sample constant on one axis and threshold constant on the other; darker or lighter cells encode the plotted metric named in the title. The default paper setting is $(c_s, c_\theta) = (2, 4)$.



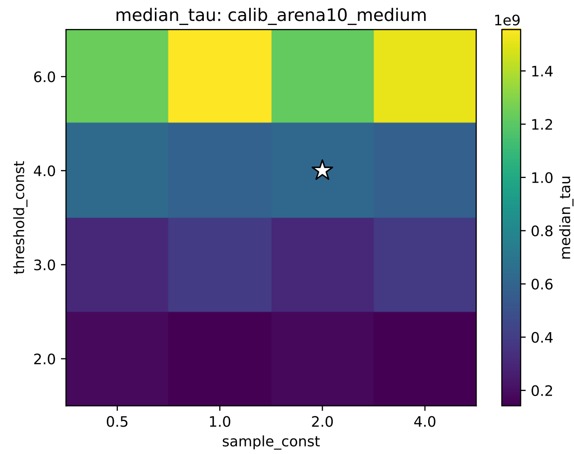
(a) Empirical error frontier

(b) Wilson upper frontier

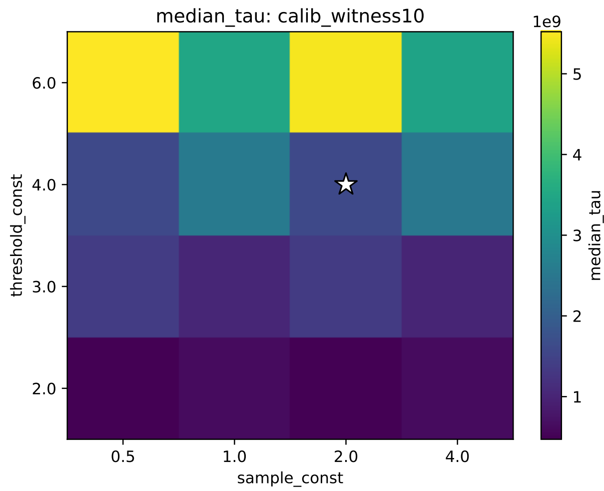
Figure 7: Constant-sensitivity error diagnostics. The x-axis is median stopping time for one VB-EGE constant pair. The y-axis is empirical error in panel (a) and the 95% Wilson upper error bound in panel (b). Every tested point has zero observed error, so these panels do not identify an efficiency–reliability frontier.



(a) Symmetric K64 d10



(b) Arena-10 medium



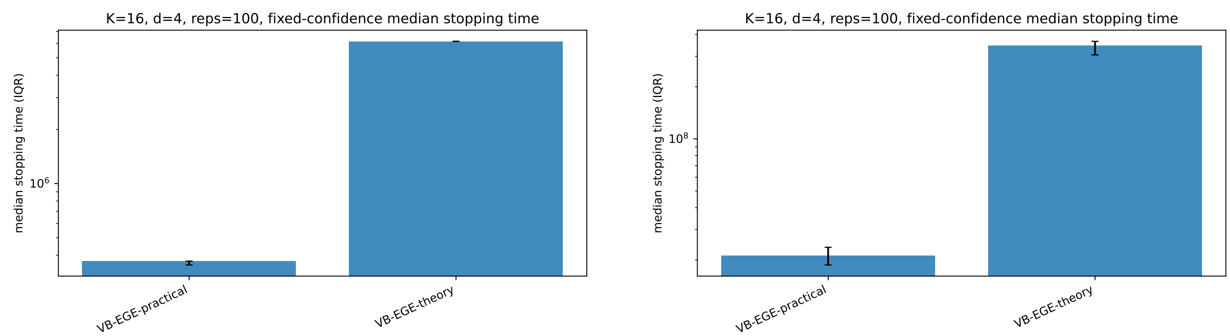
(c) Witness-10

Figure 8: Constant-sensitivity median stopping-time heatmaps. The heatmap axes are sample constant c_s and threshold constant c_θ; each cell is one VB-EGE constant pair, and the color scale is median stopping time.

Result interpretation. This grid establishes stopping-time sensitivity, not a full efficiency-reliability frontier, because all observed errors are zero. Smaller constants reduce stopping time on these instances, while the default practical constants remain the common reference point for the main comparisons.

Table 11: Theory/practical constants sanity check. Both variants use the same instances and observation replications. The theory-style constants are $(c_s, c_\theta) = (8, 16)$ and the practical constants are $(2, 4)$.

Setting	Practical median τ	Theory median τ	Theory/practical	Practical err	Theory err
Symmetric K16 d4	3.71×10^5	6.17×10^6	16.6x	0.000	0.000
Arena-4 K16	2.11×10^7	3.46×10^8	16.3x	0.000	0.000



(a) Symmetric K16 d4

(b) Arena-4 K16

Figure 9: Practical and theory-style constants on small instances. The y-axis is median stopping time with q25–q75 error bars. Both variants preserve the same qualitative behavior and zero observed error, while $(8, 16)$ is deliberately more conservative than $(2, 4)$.

6 Pareto-Size Ablation

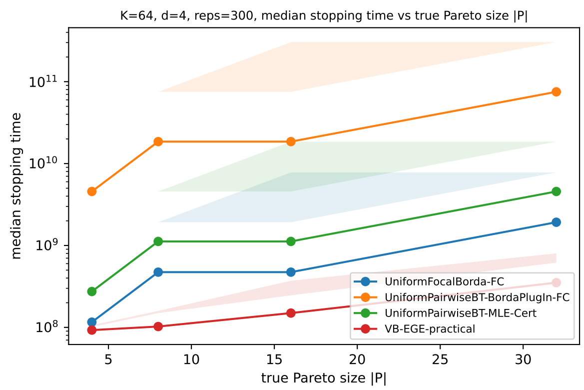
Table 12: Pareto-size ablation settings. The arena generator is held fixed while the target frontier size $s = |P|$ changes.

Setting	Generator	Grid	Purpose	Reps
Arena-4	arena_tradeoff_frontier	K=64; d=4; s in {4,8,16,32}; margins [0.08,0.25]	Measure how certificate cost changes as the frontier grows.	300
Arena-10	arena_tradeoff_frontier	K=64; d=10; s in {4,8,16,32}; margins [0.06,0.20]	Measure how certificate cost changes as the frontier grows.	300

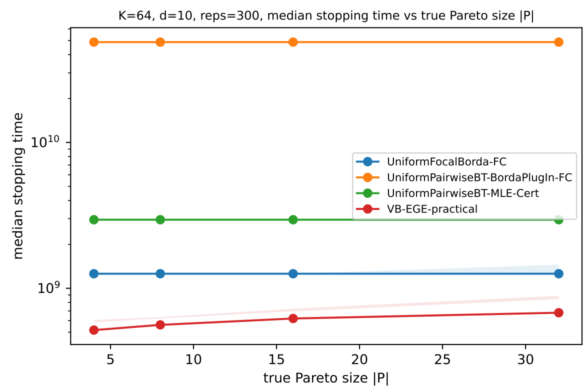
Table 13: Pareto-size ablation results. Ratios are mean stopping-time ratios relative to VB-EGE at the same arena dimension and target Pareto size.

Setting	$\|P\|$	VB $\bar{\tau}$	VB $\tau/(dH_B)$	Focal/VB	MLE/VB	Plug-in/VB
Arena-4	4	9.44×10^7	242.7	2.60x	6.11x	99.1x
Arena-4	8	1.16×10^8	240.6	10.00x	22.9x	387x
Arena-4	16	2.52×10^8	245.5	21.8x	39.7x	640x
Arena-4	32	4.42×10^8	249.0	11.8x	28.2x	462x
Arena-10	4	5.21×10^8	253.4	2.41x	5.66x	93.5x
Arena-10	8	5.55×10^8	264.2	2.27x	5.32x	87.9x
Arena-10	16	6.10×10^8	279.0	2.13x	4.99x	82.4x
Arena-10	32	6.88×10^8	285.6	2.19x	5.09x	83.7x

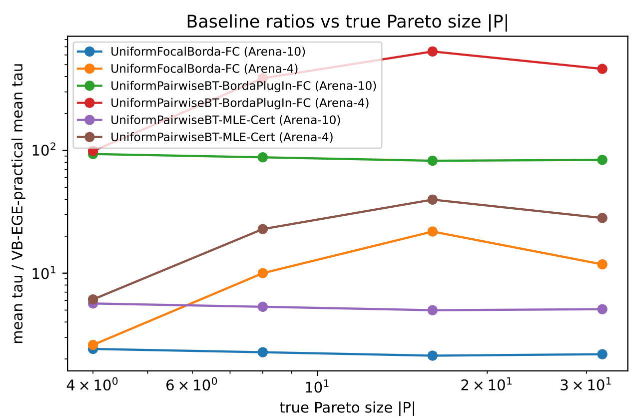
This ablation holds the arena geometry fixed while increasing the true Pareto frontier size $|P|$. In the plots, x-axis values are the target frontier sizes and each legend entry is an algorithm. The baseline-ratio plot reports mean stopping time divided by VB-EGE at the same setting; values above one favor VB-EGE.



(a) Arena-4 tau

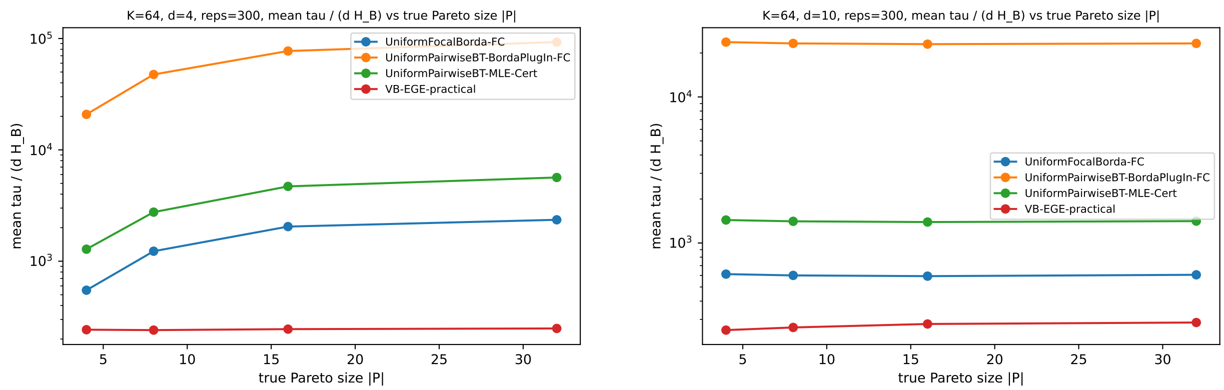


(b) Arena-10 tau



(c) Baseline/VB ratio

Figure 10: Pareto-size stopping-time ablation. The x-axis is target Pareto size $|P|$. Tau panels use median stopping time on the y-axis; the ratio panel uses $\bar{\tau}_{\text{baseline}}/\bar{\tau}_{\text{VB}}$ from cell means.



(a) Arena-4 normalized

(b) Arena-10 normalized

Figure 11: Pareto-size normalized stopping times. The x-axis is target Pareto size $|P|$ and the y-axis is mean $\tau/(dH_B)$. Normalization by dH_B separates hardness changes from pure frontier-size effects.

Result interpretation. The important question is whether the relative advantage of VB-EGE persists as the Pareto frontier grows. When the ratio curves stay above one, the adaptive Vector-Borda certificate remains more sample-efficient than uniform baselines even though more arms need to be accepted.

7 Correlated-Arena Diagnostic

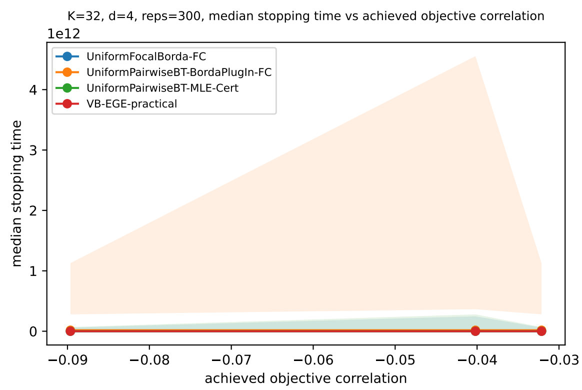
Table 14: Correlated-arena settings. The correlated generator constructs frontier anchors from shared latent factors and records the achieved average off-diagonal objective correlation.

Setting	Generator	Grid	Purpose	Reps
Correlated Arena-4	correlated_arena_like	K=32; d=4; s=8; rho in {0,0.6,0.9}	Test whether correlated objectives change sample complexity.	300
Correlated Arena-10	correlated_arena_like	K=64; d=10; s=16; rho in {0,0.3,0.6,0.85,0.95}	Test whether correlated objectives change sample complexity.	300

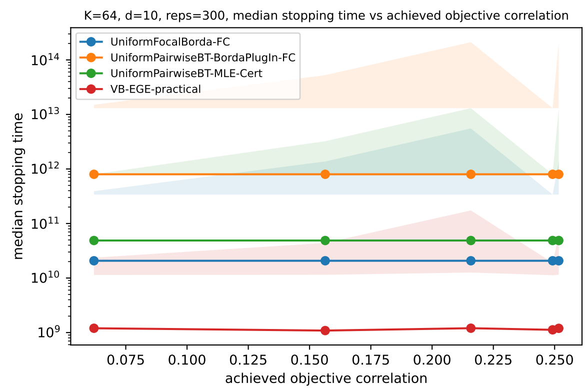
Table 15: Correlated-arena results. Achieved corr. is the mean off-diagonal objective correlation produced by the generator.

Setting	Target ρ	Achieved corr.	VB $\bar{\tau}$	VB $\tau/(dH_B)$	Focal/VB	MLE/VB	Plug-in/VB
Correlated Arena-4	0	-0.09	6.16×10^{10}	232.2	15.5x	18.2x	296x
Correlated Arena-4	0.60	-0.04	6.42×10^{10}	237.9	16.0x	18.1x	293x
Correlated Arena-4	0.90	-0.03	2.05×10^{9}	238.9	14.7x	16.2x	272x
Correlated Arena-10	0	0.06	1.46×10^{10}	258.6	30.7x	69.2x	1171x
Correlated Arena-10	0.30	0.16	1.03×10^{13}	257.8	32.0x	75.1x	1213x
Correlated Arena-10	0.60	0.22	2.54×10^{11}	273.8	28.3x	74.2x	1211x
Correlated Arena-10	0.85	0.25	4.94×10^{11}	268.5	53.7x	57.3x	2062x
Correlated Arena-10	0.95	0.25	1.01×10^{10}	274.1	54.2x	68.0x	1164x

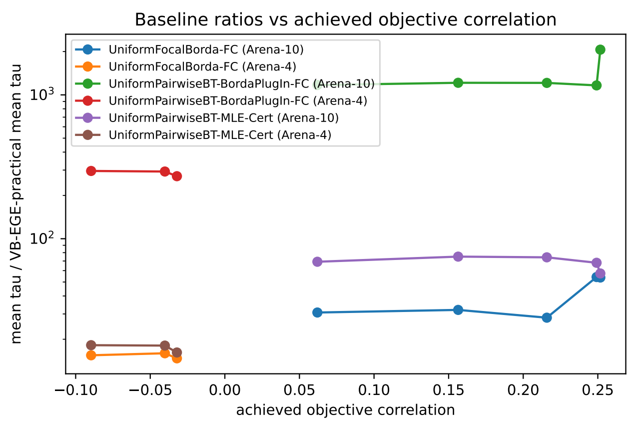
The correlated arena generator attempts to vary a target objective-correlation parameter ρ, but the achieved correlations are modest because the generator also enforces a requested Pareto-front size. Therefore these runs should be read as diagnostics rather than a strong correlation-robustness claim. The achieved-correlation column reports the actual mean off-diagonal correlation after resampling. All plotted x-axes use achieved correlation rather than the requested target. The pair-cell coverage diagnostic is meaningful only for pairwise baselines, while the Borda-gap plot reports the average minimum Borda certificate gap.



(a) Arena-4 tau

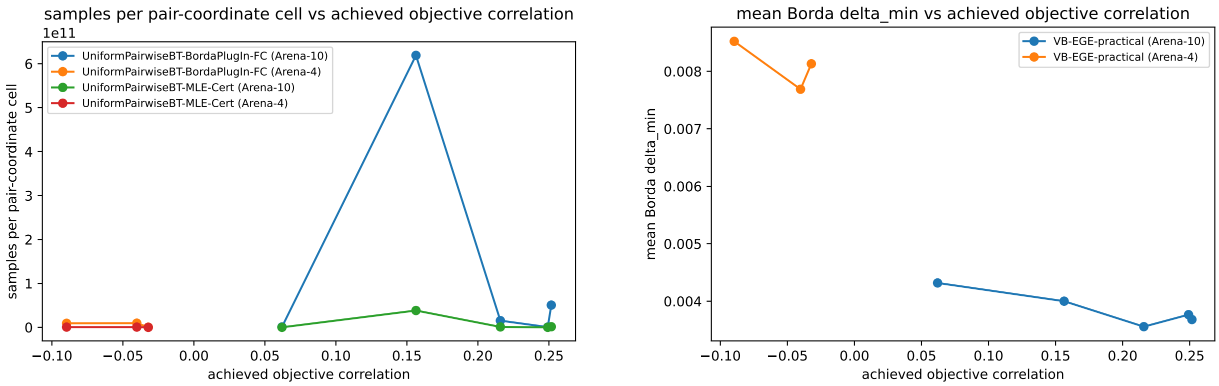


(b) Arena-10 tau



(c) Baseline/VB ratio

Figure 12: Correlated-arena stopping times. The x-axis is achieved objective correlation. Tau panels use median stopping time; the ratio panel uses the ratio of mean stopping times. Lower tau curves and higher ratios favor VB-EGE.



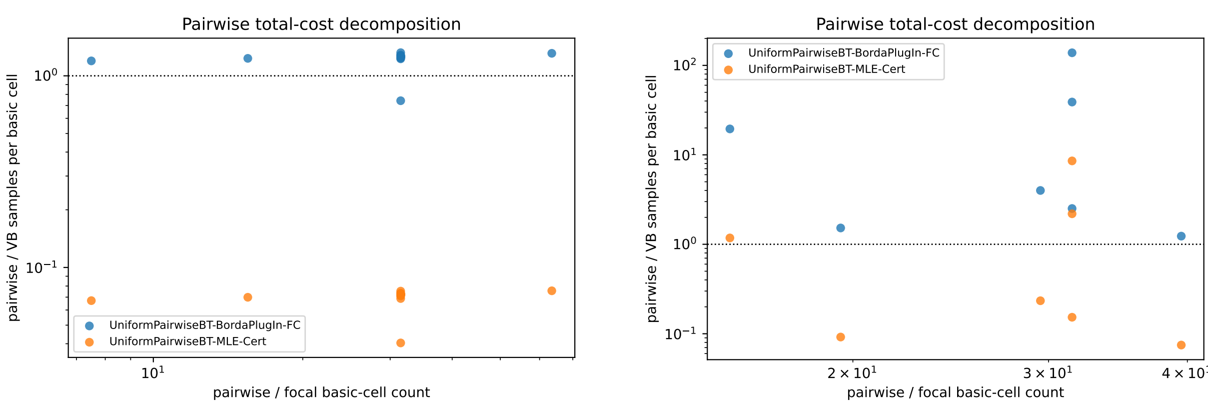
(a) Pair-cell coverage

(b) Minimum Borda gap

Figure 13: Correlated-arena diagnostics. The x-axis is achieved objective correlation. The coverage y-axis is mean samples per pair-coordinate cell τ/C for pairwise baselines, with $C = dK(K-1)/2$; the gap y-axis is mean $\Delta_{\min,B}$.

Result interpretation. Within the modest achieved-correlation range, correlation changes both the geometry of the objectives and the effective Borda gaps. If stopping time increases while the minimum Borda gap shrinks, the harder certificate explains the cost. If pairwise baselines grow faster than VB-EGE at similar Borda gaps, the extra cost comes from their broad direct pair-coordinate sampling requirement.

8 Sampling-Mechanism Diagnostics



(a) Scaling tasks

(b) Benchmark tasks

Figure 14: Pairwise total-cost decomposition. Only pairwise baselines appear. The x-axis is the structural basic-cell multiplier $C/(Kd) = (K-1)/2$. The y-axis is the per-cell sampling-burden ratio $(\tau_{\text{pair}}/C)/(\tau_{\text{VB}}/(Kd))$. Their product is the total stopping-time ratio $\tau_{\text{pair}}/\tau_{\text{VB}}$, so the two axes no longer share the same stopping time mechanically.

This decomposition separates the number of basic cells from the average evidence collected in each cell. Pairwise methods pay for both a larger cell set and, depending on the certificate, a different per-cell burden.

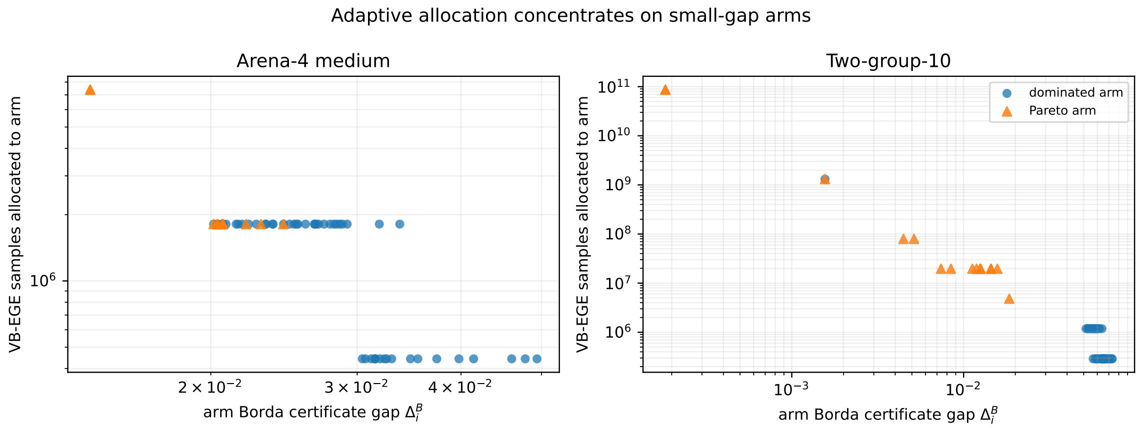


Figure 15: Arm allocation versus Borda certificate gap. Each point is one arm in a representative run. The x-axis is its true Borda certificate gap Δ_i^B and the y-axis is $\sum_r N_{ir}$, the total VB-EGE samples assigned to that arm. The log–log pattern directly shows that smaller-gap arms receive more samples.

The allocation plot is the direct adaptivity diagnostic: VB-EGE concentrates its sampling on arms close to the decision boundary instead of maintaining uniform coverage over all arms or all pair-coordinate cells.

9 Fixed-Error Budget Calibration

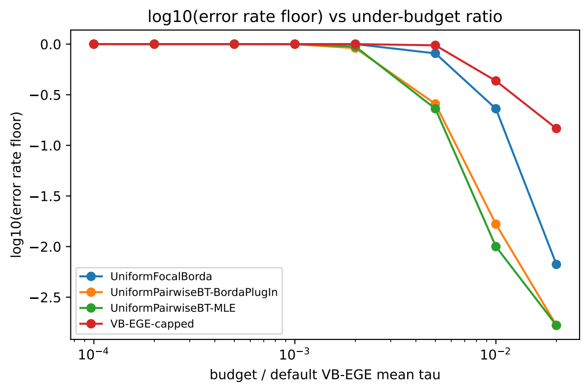
Table 16: Fixed-error budget calibration settings. Budgets are reported as fractions of the default fixed-confidence VB-EGE median stopping time on the matching benchmark task.

Setting	Generator	Parameters	Budget ratios	$\tau_{\mathrm{VB,ref}}$	Reps
Arena-10 medium	arena_tradeoff_frontier	K=64; d=10; s=16; margins [0.06,0.20]	1e-4, 2e-4, 5e-4, 1e-3, 2e-3, 5e-3, 1e-2, 2e-2	benchmark median	300
Witness-10	unique_witness_d	K=80; d=10; s=16; q=4; margins [0.035,0.14]	0.02, 0.05, 0.1, 0.2, 0.5, 1.0	1.61e9	300

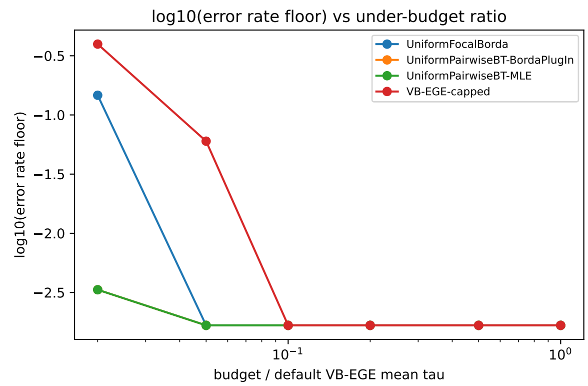
Table 17: Fixed-error budget ratios. Entries are the first tested budget ratio $T/\widetilde{\tau}_{\mathrm{VB,ref}}$ with empirical error at most 5%. Ratios compare each baseline budget ratio with VB-EGE-capped on the same setting.

Setting	VB-capped	Focal/VB	MLE/VB	Plug-in/VB
Arena-10 medium	–	–	–	–
Witness-10	0.10	0.50x	0.20x	0.20x

This calibration is not the baseline definition. The fixed-confidence baseline algorithms above remain the main comparison. Here, capped variants are run under a fixed budget to show what happens before the certification rule has enough samples. The x-axis is the budget divided by the default fixed-confidence VB-EGE median stopping time, and the table reports the first tested budget ratio reaching empirical error at most 5%.

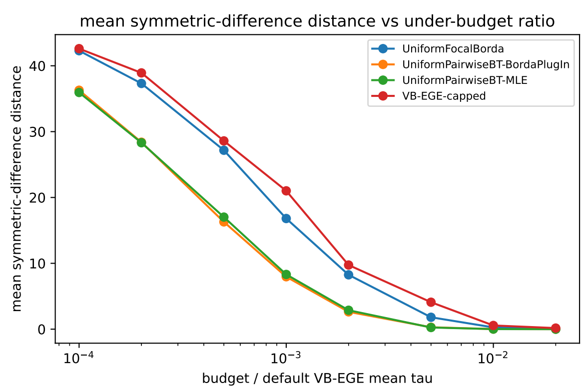


(a) Arena-10 medium

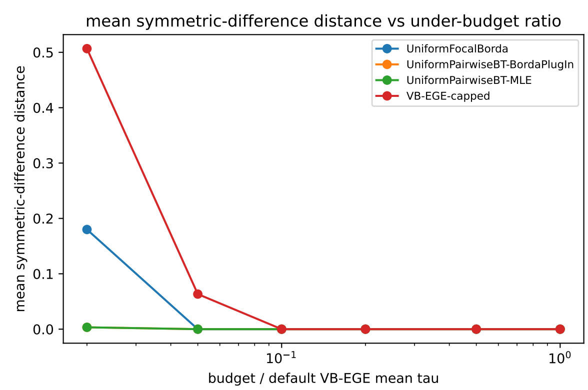


(b) Witness-10

Figure 16: Fixed-error budget calibration: empirical error. The x-axis is budget ratio $T/\widetilde{\tau}_{\mathrm{VB,ref}}$ and the y-axis is $\log_{10}$ empirical error, with the finite-sample floor used only when a method has no observed failures.



(a) Arena-10 medium



(b) Witness-10

Figure 17: Fixed-error budget calibration: Hamming distance. The x-axis is budget ratio $T/\widetilde{\tau}_{\mathrm{VB,ref}}$ and the y-axis is mean symmetric-difference distance $|\widehat{P} \triangle P|$.

The pre-certification curves are intentionally not presented as a VB-EGE win. On the revised Arena-10 grid, at budget ratio 0.02, VB-EGE-capped still has empirical error 0.147, while focal Borda has 0.0067 and both pairwise methods have zero observed error; MLE and Borda plug-in already fall below 5% at ratio 0.01. Witness-10 likewise shows that pairwise MLE and Borda plug-in can reach 5% empirical error at smaller capped budgets. This distinguishes fixed-confidence certification efficiency from arbitrary fixed-budget ranking. The expanded Arena grid probes ratios down to 10^{-4} so its error transition is visible rather than being hidden at the zero-error floor.

10 Empirical Error Evidence

All completed main fixed-confidence cells have zero observed Pareto-set errors. A zero count is summarized by the 95% Wilson upper bound in the result tables; it is not converted into a claim that finite replications verify very small target values such as $\delta = 0.001$. In particular, 500 zero-error replications give a Wilson upper bound of about 0.0076, so the confidence sweep supports stopping-time dependence on $\log(1/\delta)$ rather than direct validation of a one-in-a-thousand error rate.

11 Takeaways

1. All completed main fixed-confidence synthetic experiments achieved zero empirical Pareto-set error.
2. VB-EGE-practical loses essentially nothing to uniform focal Borda on symmetric instances and is substantially more sample-efficient on heterogeneous Pareto geometries.
3. Pairwise BT-MLE can estimate latent pairwise signs accurately, but fixed-confidence Pareto certification makes its broad pair-coordinate coverage expensive.
4. The largest gains occur in high-dimensional arena and two-group settings, where adaptive Vector-Borda certificates avoid many unnecessary pairwise comparisons.
5. Arena-10 benchmark and constant-sensitivity results now use one versioned instance bank and agree exactly at the common default configuration; confidence scaling retains a separate paired-δ bank.
6. The BT-MLE comparator is an empirical fixed-confidence-style certificate, while the fixed-error budget calibration is a separate pre-certification stress test rather than a replacement baseline.

Reproducibility Notes

The source summaries used in this report are `results/summary/fixed_confidence_scaling_summary.csv` and `results/summary/fixed_confidence_benchmarks_summary.csv`, plus `results/summary/fixed_confidence_benchmarks_summary_paired.csv`, `results/summary/confidence_scaling_quantile_summary.csv`, `results/summary/constants_calibration_summary.csv`, `results/summary/pareto_size_ablation_summary.csv`, and `results/summary/correlated_arena_summary.csv`, plus `results/summary/under_budget_stress_summary.csv` for the fixed-error budget calibration, and `results/summary/theory_constants_sanity_summary.csv`. The raw completed main outputs are `results/raw/fixed_confidence_scaling.csv`, `results/raw/fixed_confidence_benchmarks.csv`, and the corresponding CSV fallbacks under `results/raw/` for the diagnostic extensions. Smoke runs are excluded from the main conclusions because they were used only to validate the implementation path before the full runs. Revised randomized benchmarks store `instance_id`, `theta_hash`, instance index, and observation replicate in every raw row.